



5G CONCEPT AND OPPORTUNITIES FOR AFRICA

WATRA CAPACITY BUILDING WORKSHOP ON 5G POLICY AND OPPORTUNITY FOR AFRICA
23 APRIL 2019, OUAGADOUGOU

Dr. Bienvenu AGBOKPONTO SOGLO
Government and Policy Director – Africa



AGENDA

1. BRIEF INTRODUCTION TO INTEL
2. BARRIERS TO ACCESS AND PUBLIC POLICIES TO OVERCOME BARRIERS
3. 5G OPPORTUNITY FOR AFRICA
4. 5G CONCEPT

BRIEF INTRODUCTION TO INTEL

AT INTEL

WE'RE POWERING THE **FUTURE OF COMPUTING AND COMMUNICATIONS**,
DELIVERING **EXPERIENCES** ONCE THOUGHT TO BE IMPOSSIBLE.



INTEL BY THE NUMBERS

1968

Year Robert Noyce and Gordon Moore founded Intel

\$70.8B

2018 Revenue

102,700

Number of employees as of January 2018

\$13B

R&D budget in 2017

200M

Approximate number of latest-gen transistors that fit on the head of a pin

900M

Cores shipped in the past 5 years powering the world's data centers and cloud infrastructure

\$1B

Amount invested in companies that are helping to advance artificial intelligence

30%

Estimated reduction in fatal accidents in Intel-powered ADAS-equipped vehicles

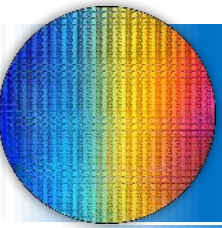
84%

Improvement in gap to full representation in Intel's U.S. workforce since 2015

3B

Kilowatt-hours of energy saved by energy conservation projects since 2012

INTEL TECHNOLOGY LEADERSHIP



**MAKE THE WORLD'S
BEST SEMICONDUCTORS**

**BE THE LEADING END TO END PLATFORM
PROVIDER FOR THE NEW DATA WORLD**



**LEAD THE ARTIFICIAL INTELLIGENCE
& AUTONOMOUS REVOLUTION**



**IF IT IS SMART AND CONNECTED,
IT'S BEST WITH INTEL**

widespread,
affordable,
high-quality

BROADBAND

promotes
data usage and PC

DELIVER THE BEST CUSTOMER EXPERIENCES ON THE PLANET



BARRIERS TO ACCESS AND PUBLIC POLICIES TO OVERCOME BARRIERS IN DEVELOPPING COUNTRIES

BARRIERS TO ACCESS AND PUBLIC POLICIES TO OVERCOME BARRIERS

- ❑ List of special requirements/barriers faced by developing countries and examples of strategies to overcome such barriers

Barrier/obstacle	Examples of strategies to overcome the barriers
1. Low levels of purchasing power in certain rural and sub-urban areas	• Subsidies to the benefit of end-users, to ensure broadband adoption, once access is secured; • Discounted offers from operators to end-users; • Telecentres for shared use to kick- start broadband markets; • Public-private partnerships (PPPs)
2. Limited financial resources available via some USFs	• Policy-makers should work with operators, depending on local needs and government funding, to ensure USF is properly sourced and effective. • Support (e.g. from international agencies) for ad-hoc projects. • Priority given to UAS projects based on strict and clear criteria
3. The low levels of ICT skills of some of the population	• ICT training; • Connecting up educational establishments • ICT lessons in schools and universities, and ICT equipment furnished at low or no cost

BARRIERS TO ACCESS AND PUBLIC POLICIES TO OVERCOME BARRIERS

- ❑ List of special requirements/barriers faced by developing countries and examples of strategies to overcome such barriers (cont.)

Barrier/obstacle	Examples of strategies to overcome the barriers
4. The lack of basic commodities (water, electricity, etc.)	• Telecentres open to the public where access to commodities is guaranteed • Wi-Fi access in public spaces where access to commodities is guaranteed
5. The limited availability of consumer electronic equipment	• Distribution of equipment directly, or subsidies for consumer electronic equipment by poor households • Review import duty regimes to ensure they are effective. • Equipment approval (supply) policies should not be too onerous or restrictive.
6. High tax rates on telecom services or equipment	• Targeted tax and import duty reductions on broadband services and devices, including removal of luxury taxes.
7. Lack of infrastructure/ high costs of deployment	• National broadband plan, including roll- out of a mutualized national backbone, as well as in-building infrastructure • Grants to operators to build out infrastructure • Sharing of infrastructure and works

BARRIERS TO ACCESS AND PUBLIC POLICIES TO OVERCOME BARRIERS

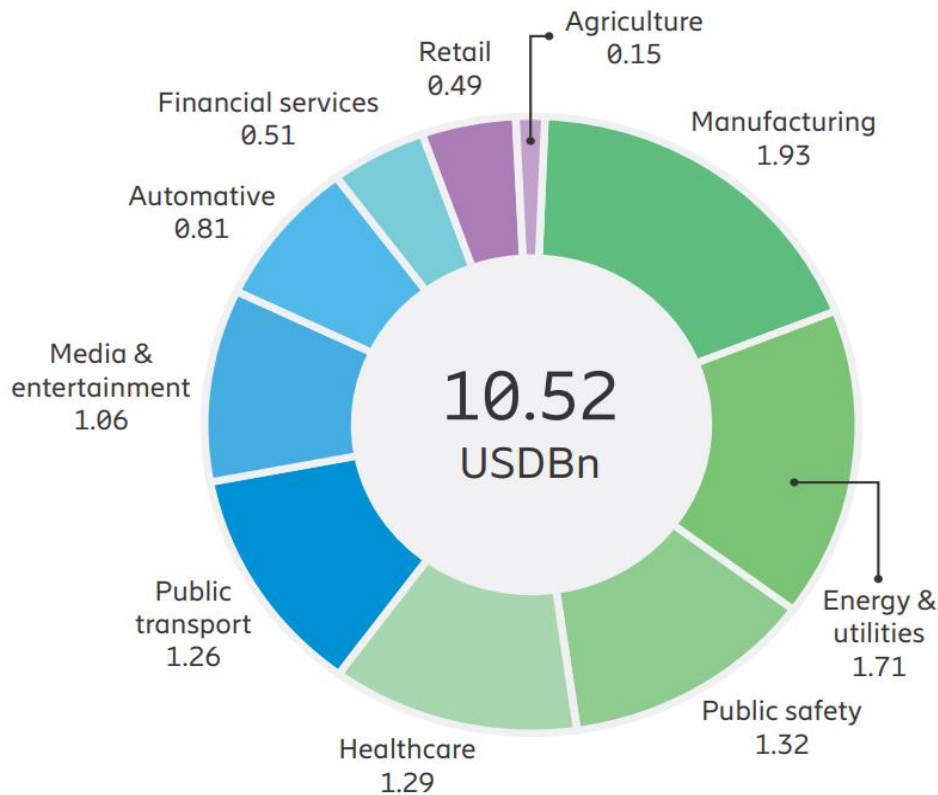
- ❑ List of special requirements/barriers faced by developing countries and examples of strategies to overcome such barriers (cont.)

Barrier/obstacle	Examples of strategies to overcome the barriers
8. Administrative delays in authorizations to deploy new infrastructure	• Involve relevant agencies and Ministries early; • Streamline licensing procedures • Eliminate red-tape and delays; • Remove barriers and obstacles to owning land
9. Limited economic growth in certain areas	• Ongoing subsidy programs on the demand side, following investment on the supply side
10. Limitations in amount of spectrum available	• Streamline spectrum licensing and re-farming practices • Implementation of the digital switch-over • More effective policies for spectrum allocation/assignment
11. Limited availability of relevant local content	• Subsidies and awards for the development of local content • Development of e-government services, open government / freedom of information policies.



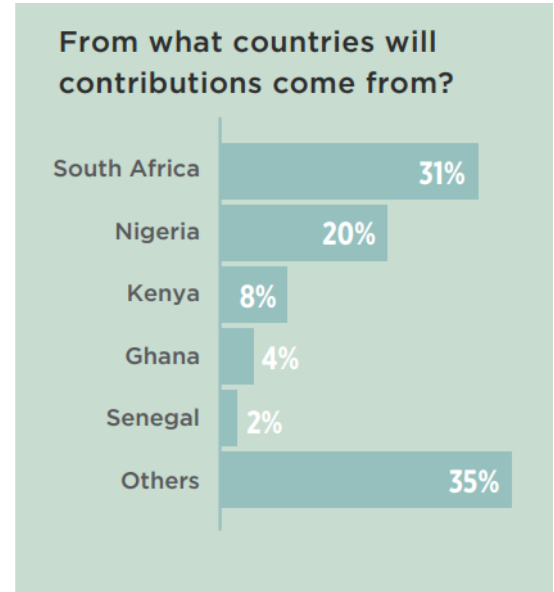
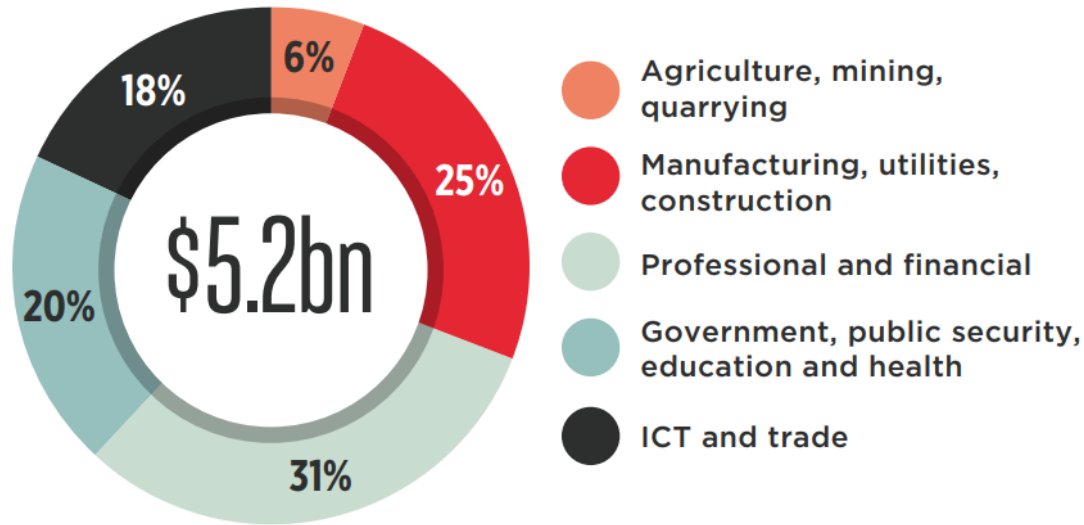
5G OPPORTUNITY FOR AFRICA

5G POTENTIAL ECONOMIC BENEFIT FOR AFRICA BY 2026¹



Estimated 5G Business Potential for Africa is in the order of **10.52 Billion USD** by **2026¹**

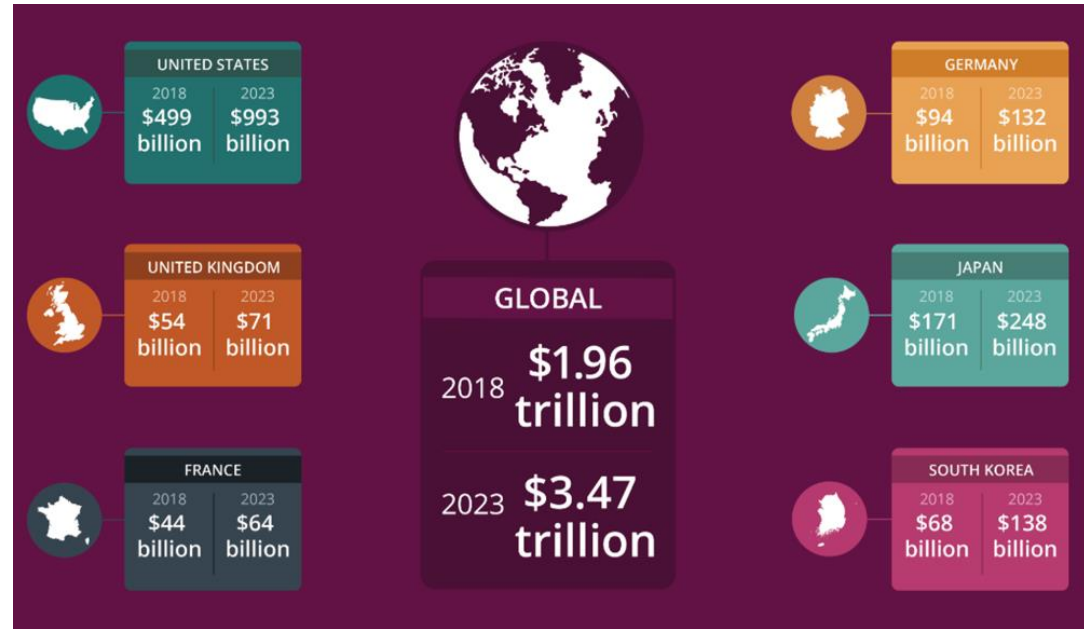
GDP IMPACT OF MMWAVE 5G IN SUB-SAHARAN AFRICA BY 2034²



❑ The GDP impact (socio-economic benefits) of mmWave 5G in Sub-Saharan Africa is **around 5.2 billion USD by 2034**

5G POTENTIAL ECONOMIC BENEFIT FOR AFRICA

- ❑ License-exempt frequency bands are also a critical component of 5G
- ❑ In 2018, the economic value provided by Wi-Fi is nearly \$2 trillion, and is expected to grow to almost \$3.5 trillion by 2023³
- ❑ Economic value is determined in terms of –
 - ✓ **Consumer surplus:** Total amount consumers are willing to pay for a product or service compared to going without it.
 - ✓ **Producer surplus:** Economic benefit producers earn from a product or service.
 - ✓ **GDP contribution:** Market value of all final goods and services produced in a period.



³ <https://www.wi-fi.org/news-events/newsroom/wi-fi-global-economic-value-reaches-196-trillion-in-2018>

5G OPPORTUNITY FOR AFRICA – BROADBAND PENETRATION

Key indicators for Sub-Sahara Africa / Africa

Key indicators	South Africa	Africa	World
Mobile subs. (%)	177.9*	82	104
Mobile Broadband sub. (%)	74.2*	43.5*	74
Fixed broadband Household Penetration (%)	< 7.5*		~ 52*

Source: Ericsson mobility report, November 2018

- ❑ Broadband subscription penetration in Sub-Sahara Africa (41%) is low compared to the world average which is 74%.

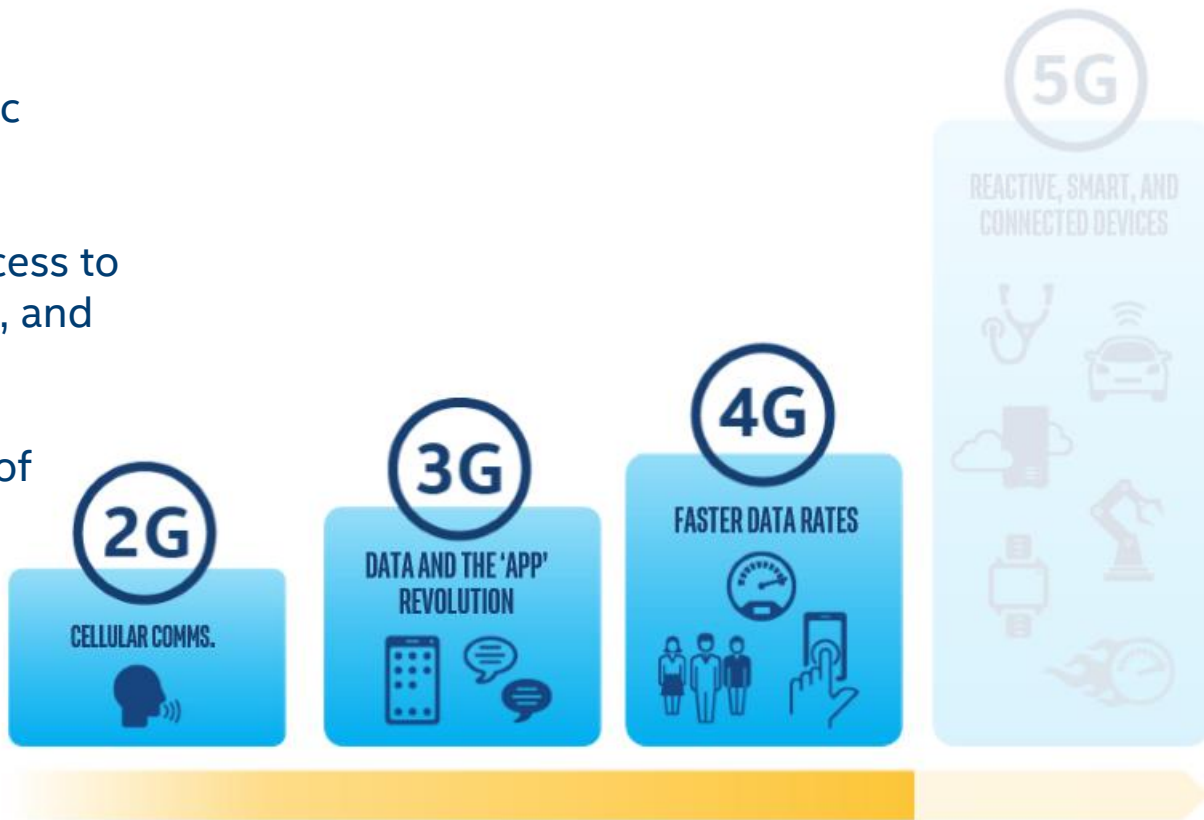


5G CONCEPT

2G - 4G WIRELESS EVOLUTION

In a mere 20 years, we've:

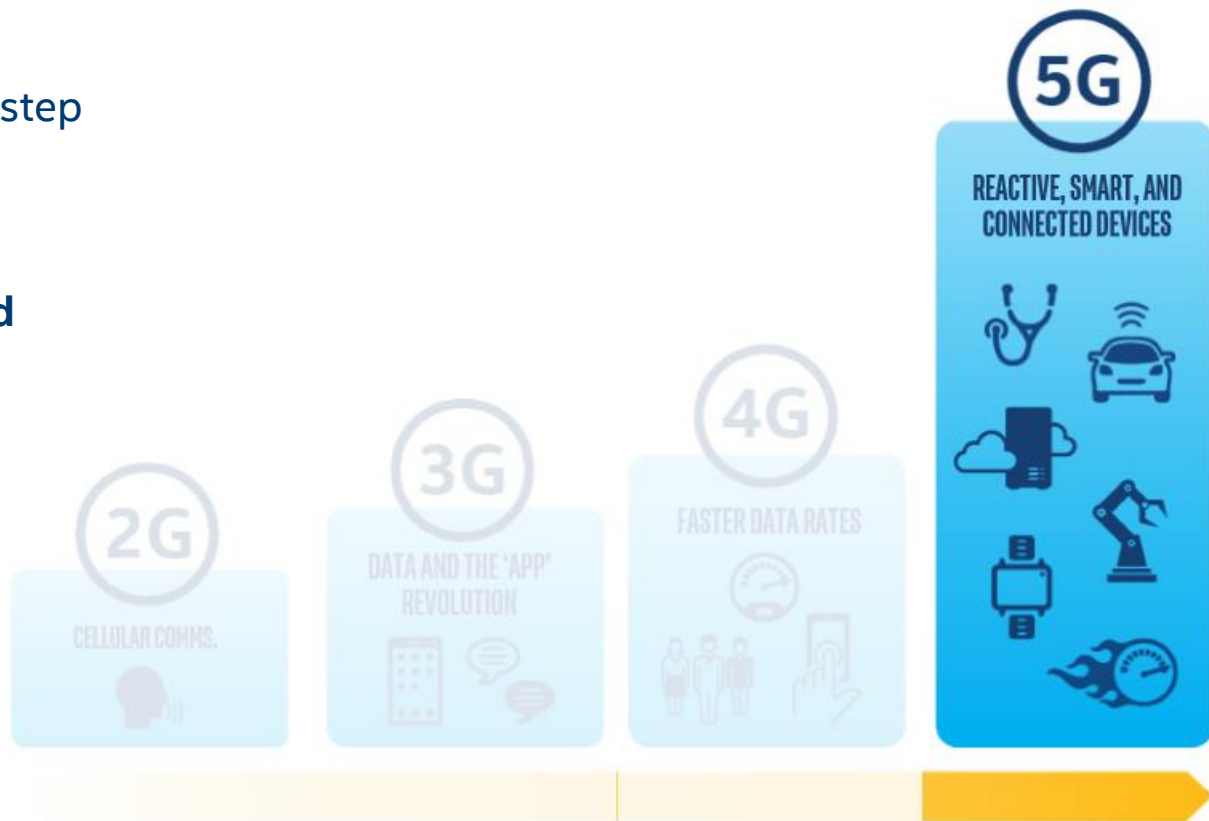
- ❑ Moved from voice-centric communications (2G)
- ❑ To IP-centric, gaining access to mobile data, information, and rich content (3G)
- ❑ At ever-increasing rates of speed (4G)



THE 5G ERA

5G: intersection of compute and connectivity

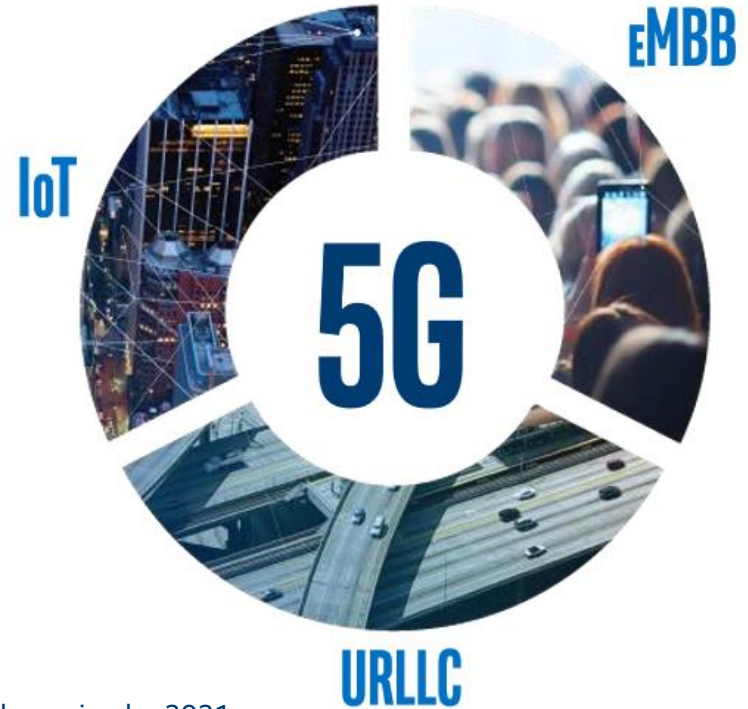
- ❑ 5G is more than just another step in the evolution of wireless.
- ❑ **5G is the convergence of wireless with computing and the cloud.**
- ❑ 5G is a new paradigm, where everything is smart, because everything is connected.



5G: BACKGROUND

5G addresses existing, emerging and future uses delivering diverse services with usage scenarios such

- ❑ Enhanced Mobile Broadband (eMBB): Immersive experiences (Virtual Reality, Augmented Reality)
- ❑ Internet of Things (IoT): applications such as mMTC (Smart home, smart cities, sensors)
- ❑ Ultra Reliable Low Latency Communications (URLLC): applications such as Mission Critical Type Communication (Autonomous driving, Industrial)

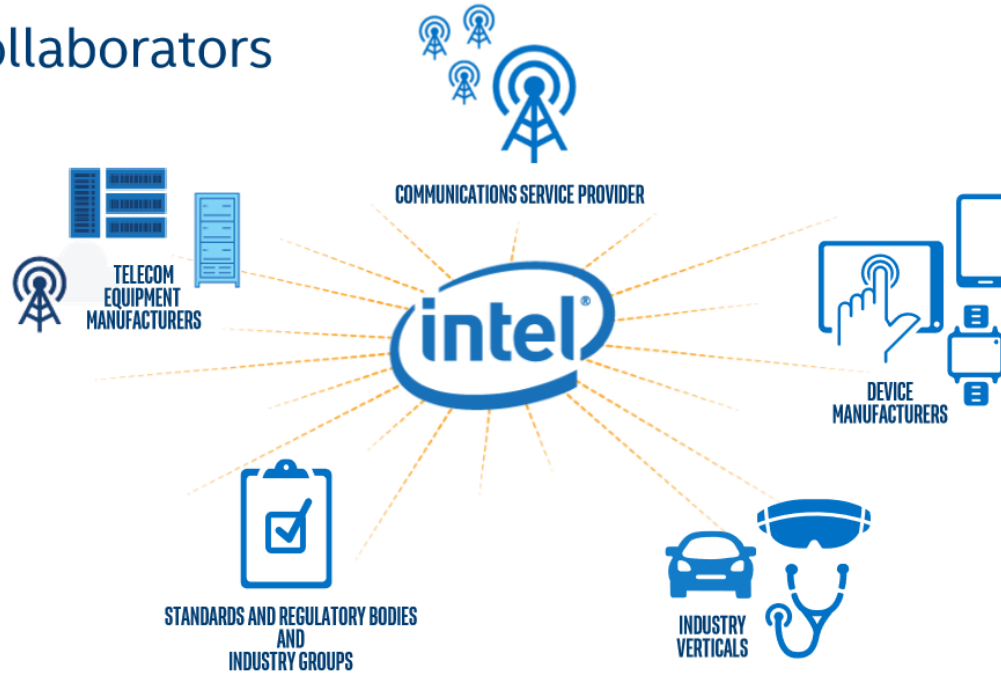


GSMA Intelligence forecasts that the first commercial 5G services to be launched in the region by 2021

5G ECOSYSTEM

- ❑ Intel participates in many standards, regulatory, and industry groups worldwide

5G Collaborators



INTEL'S 5G END TO END APPROACH

INTEL® ARCHITECTURE, FPGAS, SOFTWARE, SECURITY, AI



SCALABLE & FLEXIBLE TECHNOLOGIES AND OPEN SOURCE



Intel® Processors
& SoCs



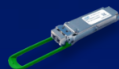
Intel® FPGA



Intel® QuickAssist
Technology



Intel® Ethernet
Adapter



Intel® Silicon
Photonics



Intel®
4G/5G Modem



CLOUD NATIVE
COMPUTING FOUNDATION



ONAP
OPEN NETWORKING AUTOMATION PLATFORM



DPDK



OPEN
DAYLIGHT



kubernetes



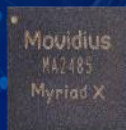
Open Source
MANO

INTEL DELIVERS END TO END PLATFORMS

DEVICES/EDGE

NETWORK

CLOUD/DATA CENTER



MEMORY

4G/5G

SOFTWARE

THE 5G ERA: 5G TECHNOLOGIES

- ❑ Intel believes that **both 3GPP-based and IEEE-based technologies** play an important role in 5G. **3GPP standards** (e.g. Release 15 and beyond, IMT-2020) and **IEEE-based standards** (Wi-Fi 6 and beyond, and WiGig) will deliver wireless broadband to consumers and businesses worldwide.
- ❑ 5G will rely on both licensed and license exempt spectrum, supported by 3GPP (NR/IMT) and IEEE (Wi-Fi and WiGig) technologies

5G TECHNOLOGIES: IMT-2020

Enhanced mobile broadband

Gigabytes in a second



3D video, UHD screens



Work and play in the cloud



Augmented reality



Industry automation



Mission critical application



Self driving car

Future IMT

Smart home/building



Voice



Smart city

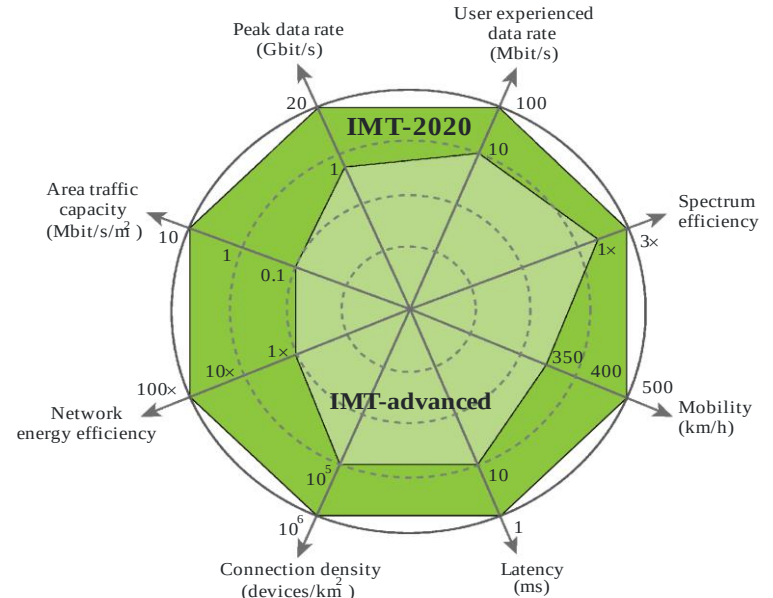


Massive machine type communications

Ultra-reliable and low latency communications

M.2083-02

Usage scenarios of IMT for 2020 and beyond



M.2083-03

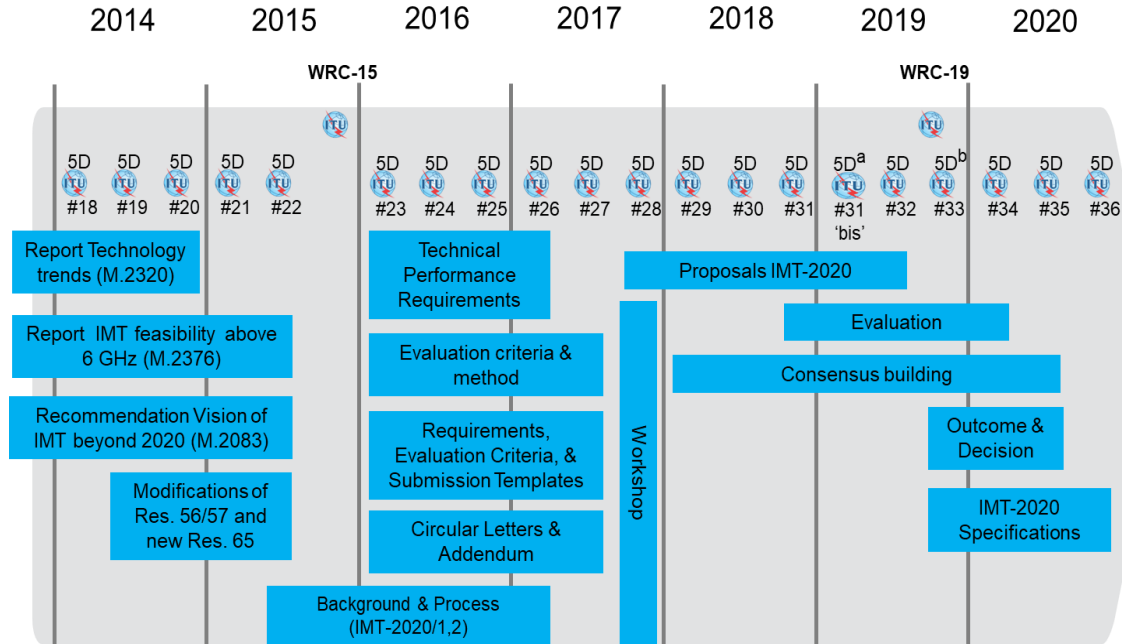
Enhancement of key capabilities from IMT-Advanced to IMT-2020

5G TECHNOLOGIES - CURRENT AND NEXT GENERATION WI-FI TECHNOLOGIES

- ❑ Wi-Fi technology is expected to play an important role in supporting 5G networks, with ultra-dense, high-speed connections to wireless and wired networks, making end-to-end communications seamless
- ❑ Wi-Fi will continue to deliver connectivity and will carry a bulk of the world's data traffic as 5G networks are deployed
 - ✓ Wi-Fi 6, based on the IEEE 802.11ax standard, will bring increased access and capabilities
- ❑ WiGig enhancements, based on the IEEE 802.11ad/ay standards and operating within the 57 – 71 GHz band, will deliver faster speeds and longer ranges for WiGig
 - ✓ 802.11ad and WiGig Technology allows up to 4 non-overlapping channels of around 2 GHz each
- ❑ According to a recent study conducted by Cisco, it is predicted that there will be 71% data offload to Wi-Fi for 5G networks by 2022; compare to 59% and 40% data offload for 4G and for 3G respectively*.

5G STANDARDIZATION - ITU

Detailed Timeline & Process for IMT-2020 in ITU-R



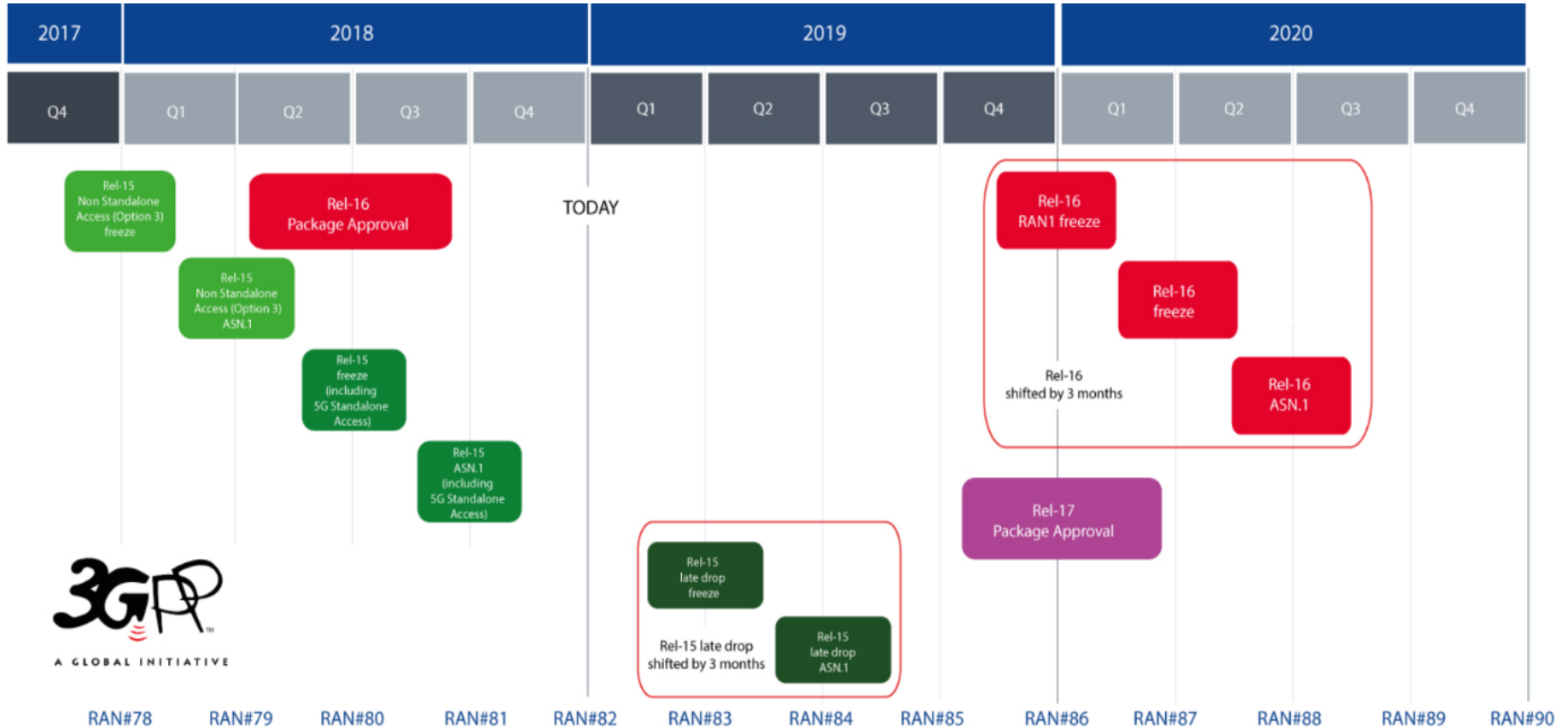
(a) – five day meeting, (b) – focus meeting on Evaluation (Technology)

Note: While not expected to change, details may be adjusted if warranted.

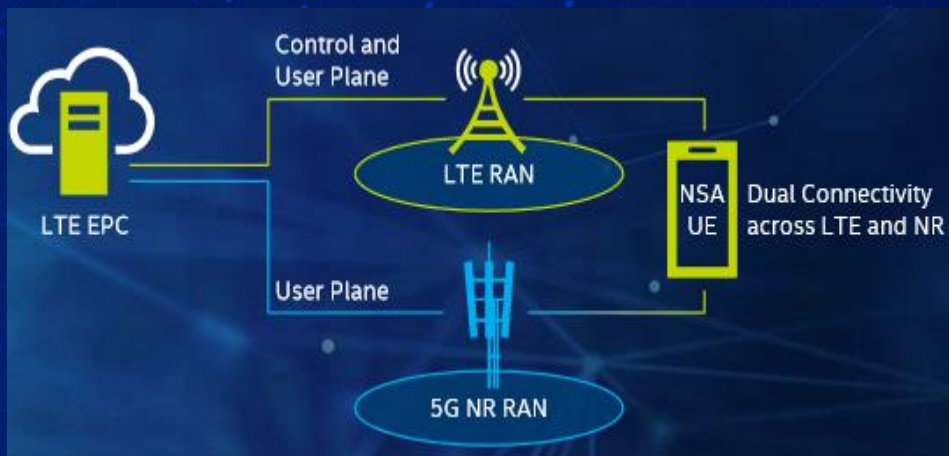
- ❑ **ITU-R M. [IMT-2020.SPECS]:** Will contain the detailed specifications of the terrestrial radio interfaces for IMT-2020; and is expected to be finalized by ITU-R Working Party 5D in October 2020)

Requirements: Report ITU-R M.2410; **Submission:** Report ITU-R M.2411; **Evaluation criteria & Method:** Report ITU-R M.2412
Evaluation process: ongoing, proposals for IMT-2020 RIT/SRIT are being submitted to ITU-R WP 5D

5G STANDARDIZATION – 3GPP



5G ARCHITECTURE OPTIONS



NON-STANDALONE (NSA)

First wave of 5G deployment

Uses 5G frequencies for improved data throughput

Leverages existing 4G deployments; smother migration to 5G



STANDALONE (SA)

Standard approved in June 2018

Simplified network infrastructure

ideal for use cases such as URLLC

DUAL CONNECTIVITY LEVERAGES EXISTING LTE DEPLOYMENTS

5G NR MAKES USE OF LTE COVERAGE FOUNDATION VIA DUAL CONNECTIVITY

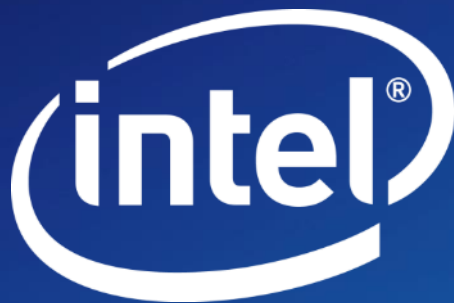
EXISTING DEPLOYMENTS

5G AUGMENTED DEPLOYMENTS



Simultaneously connected to LTE & NR base stations
Quick fall back to LTE if blockage in NR link or if UE moves out of NR Range

Thank you



bienvenu.agbokponto.soglo@intel.com